

Closed-Book Knowledge Base Construction with Calibrated Abstention and Numeric Self-Consistency

Anonymous ACL submission

Abstract

We present our system for the LM-KBC 2026 shared task on knowledge base construction from language models (5th edition, at the AKBC workshop, EMNLP 2026). Given a subject and a relation, a system must return the *complete* object set—empty, single, or up to hundreds of values—using only an open-weight model ($\leq 32\text{B}$ parameters), with no retrieval, no external knowledge, and no fine-tuning. Our central observation is that the macro metric rewards *abstention*: predicting the empty set everywhere already scores 0.175 relation-averaged macro- F_1 , near the official baseline (0.273). We therefore treat *when to answer* as a first-class decision and build a per-relation pipeline that (i) grounds prompts in each relation’s exact scope, (ii) induces calibrated abstention by self-consistency voting, and (iii) aggregates numeric relations by median. With Qwen2.5-14B-Instruct in 4-bit on a single 16GB GPU, the system reaches 0.416 relation-averaged macro- F_1 (0.447 micro), beating the official baseline on five of six relations. All generation is cached, so the entire decode stage is reproducible without a GPU. Code is released.¹

1 Introduction

Pre-trained language models (LMs) store substantial factual knowledge (Petroni et al., 2019; Roberts et al., 2020), and a line of work probes and extracts it (Jiang et al., 2020; Zhong et al., 2021; AlKhamissi et al., 2022). Turning that knowledge into reliable knowledge-base (KB) entries—the goal of KB construction and completion (Dong et al., 2014; Vrandečić and Krötzsch, 2014)—is harder than answering one factual query. The LM-KBC shared task (Alivanistos et al., 2022; Kalo et al., 2023, 2025) formalises this: given a subject s and relation r , predict the complete set of

objects $\{o_1, \dots, o_k\}$ with $k \in \{0, 1, \dots\}$. Unlike single-answer probes such as LAMA (Petroni et al., 2019), it makes no cardinality assumption and grades by string normalisation and alias matching, not entity disambiguation.

The 2026 edition is strictly *closed-book*: any open-weight model up to 32B parameters, but no web search, no retrieval, no external corpus, and no additional training. This rules out the retrieval and fine-tuning that strong prior entries relied on (Zhang et al., 2023; Asai et al., 2024), leaving parametric recall and *prompt design* as the only levers. Scoring is by macro precision, recall, and F_1 ; numeric relations use a 5% relative tolerance.

Our contributions are:

1. A metric analysis (Section 3) showing abstention is high-value: the empty prediction alone scores 0.175 relation-averaged macro- F_1 and matches or beats the official baseline on the most null-heavy relations.
2. A per-relation pipeline (Figure 1, Section 4) that grounds prompts in exact scopes and induces *calibrated abstention* via self-consistency voting, with no trained confidence model: it lifts a weak 0.5B model from 0.173 to 0.372 on the null-heavy relations and still helps a strong 14B.
3. A reproducible open-weight system on a single 16GB GPU reaching 0.416 relation-averaged macro- F_1 , beating the baseline on five of six relations, with cached generation enabling GPU-free decode-stage tuning (Sections 5–7).

2 Background and Related Work

LMs as knowledge bases. Petroni et al. (2019) cast factual recall as cloze probing; follow-ups improved prompts (Jiang et al., 2020; Shin et al., 2020; Zhong et al., 2021), measured closed-book capacity (Roberts et al., 2020), and questioned reliability (Cao et al., 2021). We target the set-valued,

¹<https://github.com/ANON/lm-kbc-2026>

alias-graded variant where the empty set is a valid answer.

The LM-KBC shared task. Prior editions (Ali-vanistos et al., 2022; Kalo et al., 2023, 2024, 2025) moved from masked-LM probing and prompt ensembling toward instruction-tuned generative systems and, in some tracks, retrieval or fine-tuning (Zhang et al., 2023). The 2026 closed-book, no-tuning, $\leq 32B$ rules motivate our parametric-recall-plus-abstention design.

Retrieval vs. closed book. Retrieval augmentation (Lewis et al., 2020; Guu et al., 2020; Asai et al., 2024) is the usual route to knowledge-intensive accuracy and helps most exactly when parametric memory is weak (Mallen et al., 2023); the 2026 rules forbid it, so we lean entirely on parametric knowledge and decoding.

Self-consistency, calibration, and abstention. Self-consistency (Wang et al., 2023; Chen et al., 2023) aggregates sampled generations to select an answer. Abstaining when uncertain is the classic reject option (Geifman and El-Yaniv, 2017) and its NLP forms—calibration (Guo et al., 2017), unanswerable QA (Rajpurkar et al., 2018), selective QA (Cole et al., 2023), and self-knowledge (Kadavath et al., 2022). We use sample disagreement as a zero-parameter confidence proxy (Kuhn et al., 2023; Manakul et al., 2023), but in a setting where the empty set is itself a high-value prediction.

3 Task and Metric Analysis

The task has six relations (Table 1): two numeric (HASCAPACITY, HASAREA), one with very large object sets (AWARDWONBY), and three string relations with frequent empty answers (COUNTRYLANDBORDERSCOUNTRY, PERSONHASCITYOFDEATH, COMPANYTRADESATSTOCKEXCHANGE). Each has a precise scope (e.g. land-only borders; city-granular death location; highest published capacity) which we copy verbatim into the prompt. We report two summaries: *relation-averaged* macro- F_1 (mean of six per-relation scores) and *micro-averaged* (over all instances).

The metric makes abstention valuable. Macro precision is 1 when no objects are predicted and recall is 1 when the gold set is empty, so an empty prediction against an empty gold set is a perfect F_1 . Predicting the

Relation	#val	empty%
COUNTRYLANDBORDERSCOUNTRY	68	25
PERSONHASCITYOFDEATH	100	45
COMPANYTRADESATSTOCKEXCHANGE	100	35
AWARDWONBY	10	0
HASCAPACITY	100	0
HASAREA	100	0

Table 1: Validation statistics. AWARDWONBY averages ~ 148 objects per subject (max 641); three string relations are often legitimately empty.

empty set for *every* validation instance therefore scores the “Empty” bars in Figure 3, averaging 0.175 over relations (0.203 micro)—on par with the official baseline’s relation-averaged 0.273 and *above* it on the two most null-heavy relations (PERSONHASCITYOFDEATH 0.45 vs. 0.16; COMPANYTRADESATSTOCKEXCHANGE 0.35 vs. 0.32). Gains thus come from recalling the numeric values and from knowing when *not* to answer.

4 System

Architecture overview. The pipeline (Figure 1) maps each (s, r) to one chat prompt, decodes one or more samples, and aggregates them per relation kind into an object set. It has no trainable parameters and targets the two analysis levers: abstention and numeric recall.

Scope-grounded prompts. Each relation’s system message states the task, the *exact* ground-truth scope (copied from the official definitions), and the answer shape; the model emits one JSON object $\{\text{"answers": [...]}\}$ (Figure 2). This makes the empty case unambiguous. Few-shot exemplars come from the training split; for null-heavy relations we interleave empty and non-empty exemplars so abstention is demonstrated, not merely instructed. Even a 0.5B model emits valid JSON on 100% of instances, so format is never the bottleneck.

Calibrated abstention via self-consistency. A bare instruction to “answer empty when unsure” is unreliable: an unconstrained small model names a death-city for living people and invents stock exchanges, dropping precision *below* the empty baseline (Section 7). Instead, for the three null-heavy relations we sample $N=3$ completions at temperature $\tau=0.7$ and keep an object only if it appears in at least a fraction θ of samples; single-valued

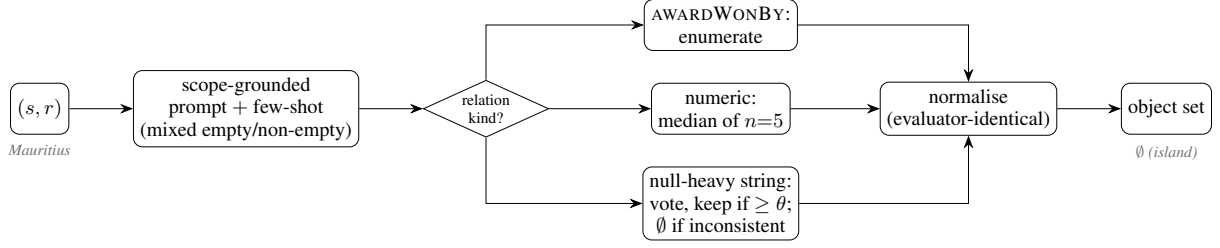


Figure 1: System pipeline. Each (s, r) maps to a scope-grounded chat prompt with mixed empty/non-empty few-shot exemplars; we decode N cached samples and aggregate by relation kind—*median* for numeric, consistency *voting with abstention* for null-heavy strings, and *enumeration* for AWARDWONBY—then normalise identically to the evaluator. The worked example (MAURITIUS, an island) yields the empty set. No trainable parameters.

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SYS: precise KB engine (closed book).
TASK: city where the subject died.
SCOPE: city granularity; [] if
alive/unknown.
Reply only {"answers":[...]}.
U: Egbert Mulder
A: {"answers":["Groningen"]}
U: Dave Keon
A: {"answers":[]}
U: <query> A:

```

Figure 2: Abbreviated prompt for a null-heavy relation. Mixed exemplars teach abstention; output is forced JSON.

Algorithm 1 CALIBRATEDABSTENTION

Require: samples $S = \{s_1, \dots, s_N\}$, threshold θ , *multi*

- 1: $c[o] \leftarrow \#\{s_i : o \in \text{PARSE}(s_i)\}$ for each object o
- 2: **if** *multi* **then return** $\{o : c[o]/N \geq \theta\}$
- 3: **else**
- 4: $o^* \leftarrow \arg \max_o c[o]$; **return** $c[o^*]/N \geq \theta ? \{o^*\} : \emptyset$
- 5: **end if**

relations abstain entirely if even the top answer is below θ (Algorithm 1). When the model guesses, samples disagree and the prediction collapses to $\emptyset$ —where the metric rewards silence. This is disabled for AWARDWONBY (recall-bound) and the numeric relations.

Numeric aggregation. HASCAPACITY and HASAREA take a single number graded within 5%. We sample $n=5$ completions and take the *median* of the parsed numbers (Wang et al., 2023) as a robustness measure.

Robust parsing. A relation-aware parser recovers the answer list from noisy output (code fences, prose), strips explicit empty markers, extracts one number for numeric relations (handling commas, units, scale words), and de-duplicates. Normalisation (lower-casing, diacritic/punctuation removal) is byte-identical to the official evaluator; pushing gold answers back through the parser scores 1.000,

confirming it is lossless.

5 Experimental Setup

Model. We use **Qwen2.5-14B-Instruct** (Qwen Team, 2024) (14.7B parameters, within budget). Several nominally “32B” models exceed the cap (Qwen2.5-32B at 32.5B; Qwen3-32B at 32.8B) and are disqualified; 14.7B was the largest model we could fit in 4-bit on our single 16 GB GPU.

Inference and reproducibility. Inference runs on one NVIDIA T4 (16 GB) via 4-bit bitsandbytes (nf4), float16 compute (Dettmers et al., 2023). We use 8-shot prompts (Brown et al., 2020) (2-shot for AWARDWONBY); the submitted system uses $\theta=0.5$, $N=3$, $n=5$, $\tau=0.7$. Generation writes a cache of raw samples; all post-processing—parsing, voting, median, threshold—is a deterministic GPU-free function of that cache, so every ablation below comes from one generation pass and is reproducible without a GPU.

6 Results

The system scores **0.416** relation-averaged macro- F_1 (**0.447** micro), versus 0.175/0.203 (empty) and 0.273/0.284 (official baseline)—a same-axis gain on both. Figure 3 and Table 2 give the per-relation picture: we beat the baseline on five of six relations, the exception being HASCAPACITY (0.09 vs. 0.10), where exact capacities are essentially unknown to the model. The strongest relations are COUNTRYLANDBORDERSCOUNTRY (0.900) and COMPANYTRADESATSTOCKEXCHANGE (0.616); the high precision on PERSONHASCITYOFDEATH ($P=0.81$) is the signature of abstention.

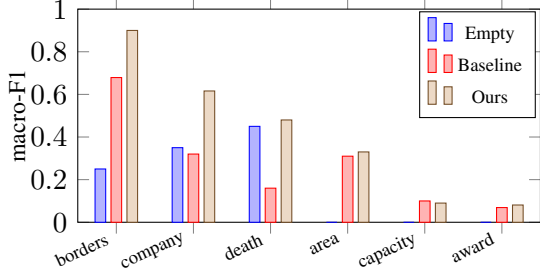


Figure 3: Per-relation validation macro-F₁: predict-nothing baseline, official baseline, and ours. We beat the official baseline on five of six relations.

Relation	Empty	Base	P	R	F1
countryLandBorders	0.250	0.679	0.91	0.95	0.900
companyTrades	0.350	0.320	0.71	0.74	0.616
personCityOfDeath	0.450	0.160	0.81	0.53	0.480
hasArea	0.000	0.310	0.33	0.33	0.330
hasCapacity	0.000	0.100	0.09	0.09	0.090
awardWonBy	0.000	0.069	0.36	0.05	0.081
Avg. of relations	0.175	0.273	–	–	0.416
Micro (all inst.)	0.203	0.284	–	–	0.447

Table 2: Validation macro-F₁. “Empty” is the measured predict-nothing baseline; “Base” the official baseline. P/R/F1 are ours ($\theta=0.5$). Bold marks relations where we beat the baseline.

7 Analysis

All ablations are decoded from the same cached samples.

Abstention rescues weak models, refines strong ones. Table 3 isolates abstention on the three null-heavy relations. On a poorly-calibrated 0.5B model the effect is dramatic ($0.173 \rightarrow 0.372$)—without voting it over-answers and falls *below* the empty baseline (0.350). The 14B is already above empty without voting (0.652), so voting adds less ($0.652 \rightarrow 0.666$) but remains positive. Abstention buys precision: on PERSONHASCITYOFDEATH it moves $P=0.77 \rightarrow 0.81 \rightarrow 0.89$ as θ rises, trading a little recall—it suppresses wrong guesses, not correct answers.

Threshold tracks the empty rate; numeric voting is neutral. Figure 4 (left) sweeps θ : the most null-heavy relation, PERSONHASCITYOFDEATH (45% empty), peaks at a *lower* θ , while the others tolerate aggressive voting—arguing for per-relation thresholds. Figure 4 (right) varies the number of numeric samples: median aggregation is non-monotonic on HASAREA and actually slightly *hurts* HASCAPACITY (single sample

	Off	$\theta=0.5$	$\theta=0.67$
0.5B (avg of 3 null-heavy)	0.173	0.363	0.372
14B (avg of 3 null-heavy)	0.652	0.665	0.666

Table 3: Self-consistency abstention, relation-averaged over the three null-heavy relations only. “Off” decodes a single sample.

0.10 vs. median-5 0.09). Decode variance is not the bottleneck—the gap is knowledge-bound; the model’s capacity guesses are the right order of magnitude but outside the 5% band (e.g. Spartan Stadium: predicted 37,000, gold 75,005).

AWAROWONBY is recall-capped. Gold sets average ~ 148 recipients (max 641) but the 14B enumerates only ~ 11 names—its outputs are *not* truncated (it closes the JSON early)—capping recall near $11/148 \approx 0.07$. Listed names are often plausible-but-wrong ($P=0.36$). This relation is bounded by parametric recall and willingness to enumerate, where a larger in-budget model or multi-step decomposition would help most.

8 Conclusion

Our system’s design follows from a metric analysis: abstention is valuable, numeric recall is scarce, and scope grounding curbs over- and under-prediction. With a single 14.7B open-weight model on commodity hardware it reaches 0.416 relation-averaged macro-F₁ (0.447 micro), beating the official baseline on five of six relations, and is fully reproducible from a cached generation pass.

Limitations

The voting threshold θ is swept on the same validation split it is evaluated on, so per-relation optima are optimistic and not held-out; we report no test numbers yet. Several margins are within plausible decoding noise: AWAROWONBY has only $n=10$ subjects, and we did not measure run-to-run variance of the stochastic decoding ($N=3$, $n=5$), so sub-0.03 deltas (e.g. the PERSONHASCITYOFDEATH +0.03) are indicative only. Being closed-book, the system inherits the base model’s coverage and recency limits and cannot recover facts it never stored, which caps AWAROWONBY recall and rare-venue numeric accuracy; a larger in-budget model (up to 32B) would likely help the numeric relations. Finally, the abstention advantage is a property of this task’s macro metric and

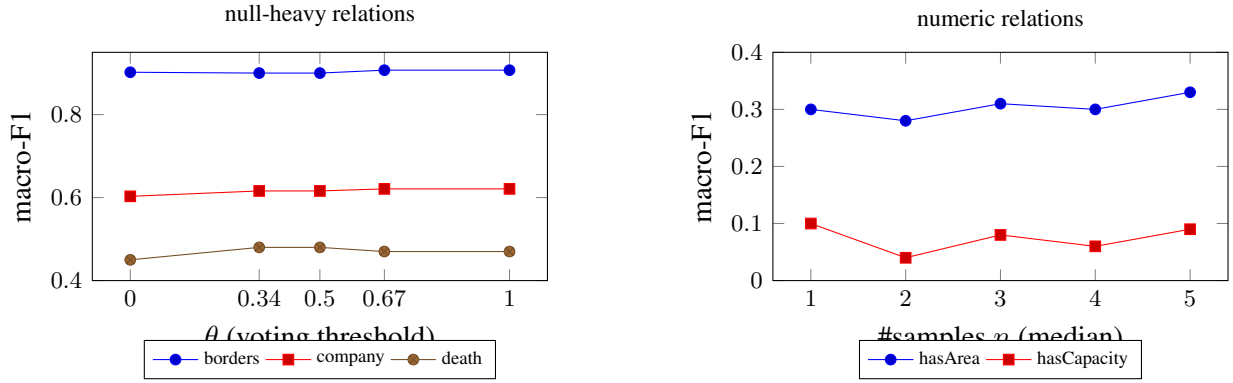


Figure 4: Left: per-relation F_1 vs. voting threshold θ (14B); the most null-heavy relation (DEATH, 45% empty) peaks at a lower θ than the others. Right: numeric F_1 vs. number of median samples; aggregation is roughly neutral, indicating a knowledge-bound gap.

the empty-as-valid-answer setting, and may not transfer to single-answer or micro-only settings.

Ethics and Generative-AI Declaration

Closed-book KB construction surfaces facts from model parameters that may be stale, biased, or hallucinated; predictions about people (e.g. place of death) can be wrong and should not be treated as authoritative without verification. The abstention mechanism partly mitigates this by emitting an empty answer rather than a confident fabrication when the model is uncertain. The only generative-AI component is the open-weight LM used for the predictions themselves, as the task requires; no proprietary assistant was used to produce results.

References

- Dimitrios Alivanistos, Selene Báez Santamaría, Michael Cochez, Jan-Christoph Kalo, Emile van Krieken, and Thiviyan Thanapalasingam. 2022. Prompting as probing: Using language models for knowledge base construction. In *Semantic Web Challenge on KB Construction from Pre-trained Language Models (LM-KBC)*.
- Badr AlKhamissi, Millicent Li, Asli Celikyilmaz, Mona Diab, and Marjan Ghazvininejad. 2022. A review on language models as knowledge bases. *arXiv preprint arXiv:2204.06031*.
- Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. 2024. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. In *ICLR*.
- Tom B. Brown and 1 others. 2020. Language models are few-shot learners. *NeurIPS*.
- Boxi Cao, Hongyu Lin, Xianpei Han, Le Sun, Lingyong Yan, Meng Liao, Tong Xue, and Jin Xu. 2021.

Knowledgeable or educated guess? revisiting language models as knowledge bases. In *ACL*.

Xinyun Chen, Renat Aksitov, Uri Alon, Jie Ren, Kefan Xiao, Pengcheng Yin, Sushant Prakash, Charles Sutton, Xuezhi Wang, and Denny Zhou. 2023. Universal self-consistency for large language model generation. *arXiv preprint arXiv:2311.17311*.

Jeremy R. Cole, Michael J.Q. Zhang, Daniel Gillick, Julian Martin Eisenschlos, Bhuwan Dhingra, and Jacob Eisenstein. 2023. Selectively answering ambiguous questions. In *EMNLP*.

Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. QLoRA: Efficient finetuning of quantized LLMs. *NeurIPS*.

Xin Dong, Evgeniy Gabrilovich, Jeremy Heitz, Wilko Horn, Ni Lao, Kevin Murphy, Thomas Strohmman, Shaohua Sun, and Wei Zhang. 2014. Knowledge vault: A web-scale approach to probabilistic knowledge fusion. In *KDD*.

Yonatan Geifman and Ran El-Yaniv. 2017. Selective classification for deep neural networks. In *NeurIPS*.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. 2017. On calibration of modern neural networks. In *ICML*.

Kelvin Guu, Kenton Lee, Zora Tung, Panupong Paspapat, and Ming-Wei Chang. 2020. REALM: Retrieval-augmented language model pre-training. In *ICML*.

Zhengbao Jiang, Frank F. Xu, Jun Araki, and Graham Neubig. 2020. How can we know what language models know? *Transactions of the ACL*.

Saurav Kadavath, Tom Conerly, Amanda Askell, and 1 others. 2022. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*.

Jan-Christoph Kalo, Tuan-Phong Nguyen, Simon Razniewski, and Bohui Zhang. 2024. Preface:

354	Knowledge base construction from pre-trained lan-	Bohui Zhang, Ioannis Reklos, Nitisha Jain, Albert	409
355	guage models (lm-kbc) 2024. In <i>LM-KBC Chal-</i>	Meroño-Peñuela, and Elena Simperl. 2023. Using	410
356	<i>lenge at ISWC, CEUR Workshop Proceedings.</i>	large language models for knowledge engineering	411
357	Jan-Christoph Kalo, Simon Razniewski, Bohui Zhang,	(LLMKE): A case study on wikidata. <i>arXiv preprint</i>	412
358	and Tuan-Phong Nguyen. 2025. LM-KBC 2025:	<i>arXiv:2309.08491.</i>	413
359	4th challenge on knowledge base construction from	Zexuan Zhong, Dan Friedman, and Danqi Chen. 2021.	414
360	pre-trained language models. In <i>LM-KBC Chal-</i>	Factual probing is [MASK]: Learning vs. learning to	415
361	<i>lenge, CEUR Workshop Proceedings.</i>	recall. In <i>NAACL.</i>	416
362	Jan-Christoph Kalo, Sneha Singhania, Simon		
363	Razniewski, and Jeff Z. Pan. 2023. LM-KBC		
364	2023: The 2nd challenge on knowledge base		
365	construction from pre-trained language models. In		
366	<i>Knowledge Base Construction from Pre-trained</i>		
367	<i>Language Models (LM-KBC), CEUR Workshop</i>		
368	<i>Proceedings.</i>		
369	Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. 2023.		
370	Semantic uncertainty: Linguistic invariances for un-		
371	certainty estimation in natural language generation.		
372	In <i>ICLR.</i>		
373	Patrick Lewis and 1 others. 2020. Retrieval-		
374	augmented generation for knowledge-intensive NLP		
375	tasks. <i>NeurIPS.</i>		
376	Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das,		
377	Daniel Khashabi, and Hannaneh Hajishirzi. 2023.		
378	When not to trust language models: Investigating		
379	the effectiveness of parametric and non-parametric		
380	memories. In <i>ACL.</i>		
381	Potsawee Manakul, Adian Liusie, and Mark J. F. Gales.		
382	2023. SelfCheckGPT: Zero-resource black-box hal-		
383	lucination detection for generative large language		
384	models. In <i>EMNLP.</i>		
385	Fabio Petroni, Tim Rocktäschel, Sebastian Riedel,		
386	Patrick Lewis, Anton Bakhtin, Yuxiang Wu, and		
387	Alexander Miller. 2019. Language models as knowl-		
388	edge bases? In <i>EMNLP-IJCNLP.</i>		
389	Qwen Team. 2024. Qwen2.5 technical report. <i>arXiv</i>		
390	<i>preprint arXiv:2412.15115.</i>		
391	Pranav Rajpurkar, Robin Jia, and Percy Liang. 2018.		
392	Know what you don’t know: Unanswerable ques-		
393	tions for SQuAD. In <i>ACL.</i>		
394	Adam Roberts, Colin Raffel, and Noam Shazeer. 2020.		
395	How much knowledge can you pack into the param-		
396	eters of a language model? In <i>EMNLP.</i>		
397	Taylor Shin, Yasaman Razeghi, Robert L. Logan IV,		
398	Eric Wallace, and Sameer Singh. 2020. AutoPrompt:		
399	Eliciting knowledge from language models with au-		
400	tomatically generated prompts. In <i>EMNLP.</i>		
401	Denny Vrandečić and Markus Krötzsch. 2014. Wiki-		
402	data: A free collaborative knowledgebase. <i>Commu-</i>		
403	<i>nications of the ACM, 57(10).</i>		
404	Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc		
405	Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery,		
406	and Denny Zhou. 2023. Self-consistency improves		
407	chain of thought reasoning in language models. In		
408	<i>ICLR.</i>		